20-2:-

⇒ Q2:-

Evaluate the followings:-

i, $(1+\omega^2)^4$

Sol

we know that the

$1+\omega+\omega^2=0$

$1+\omega=-\omega^2$

$1+\omega^2=-\omega$

$\omega+\omega^2=-1$ and $\omega^3=1$ and Note that $(\omega^3)^n$ any power equals to one.

so there fore. becase $\omega^3=1$ there fore $(1)^3=1$.

$(1+\omega^2)^4$

so $(-\omega)^4$

$=(-\omega)(-\omega)(-\omega)(-\omega)$

$=\omega^4=\omega^3\cdot\omega=\omega=1$ so

$1\times\omega$

Ans: